

Faculty of Engineering



**UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS**
Acreditación Institucional de Alta Calidad

Computer Networking

Workshop No. 2 — Sockets & Services

Systems Engineering

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Activity development

Full topology

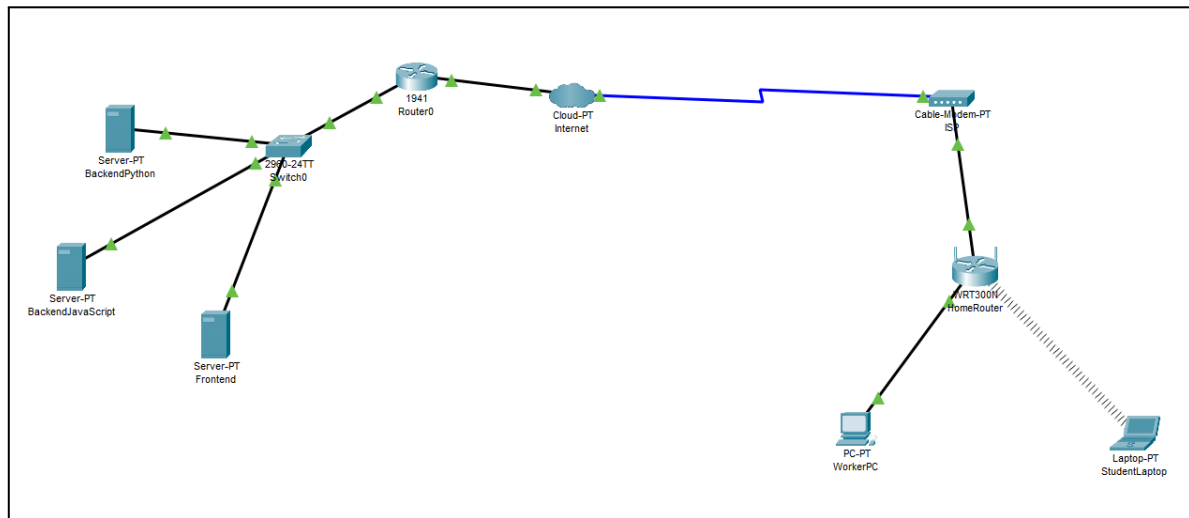


Figure 1 - Full Topology

Communication between the server and the cloud is over FastEthernet, so a **copper straight-through** cable is used.

In Packet Tracer, the Cloud PT device simulates the connection to the Internet or the Internet Service Provider (ISP) network. Coaxial cable is used to transmit data from the ISP's network to the customer's modem because it can carry *high-frequency signals* with less loss than other types of cable in terms of **price-performance ratio** (fiber is better, but much more expensive), making coaxial cable ideal for long-distance, high-bandwidth transmissions. It was necessary to install a 2.4 GHz wireless card in the laptop and configure the same network in the home router and laptop to get a connection.

Servers

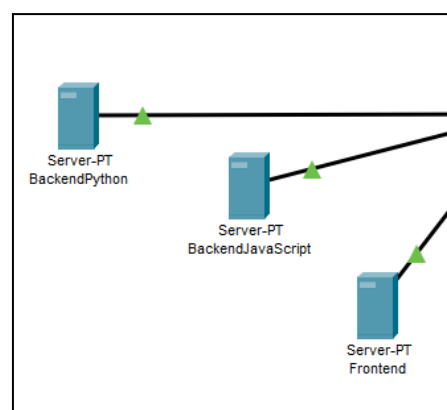


Figure 2 - Backend & Frontend Servers

The Python code creates a web server in Packet Tracer that listens for HTTP requests on port 80. When a client makes a GET request to the **"/healthcheck"** endpoint, the server responds with a verification message saying "This is a verification about python services". The `on_route_networks` function acts as a request handler, while the `main` function sets the server route and starts the server.

Communication Test

It is validated that the communication between the servers is correct and that they only work between the internal network (only between the switch), so that only the front-end server is exposed.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Serve...	BackendP...	ICMP		0.000	N	0	(edit)	
	Successful	Back...	BackendJ...	ICMP		0.000	N	1	(edit)	
	Successful	Back...	Frontend	ICMP		0.000	N	2	(edit)	

Figure 3 - ICMP Test

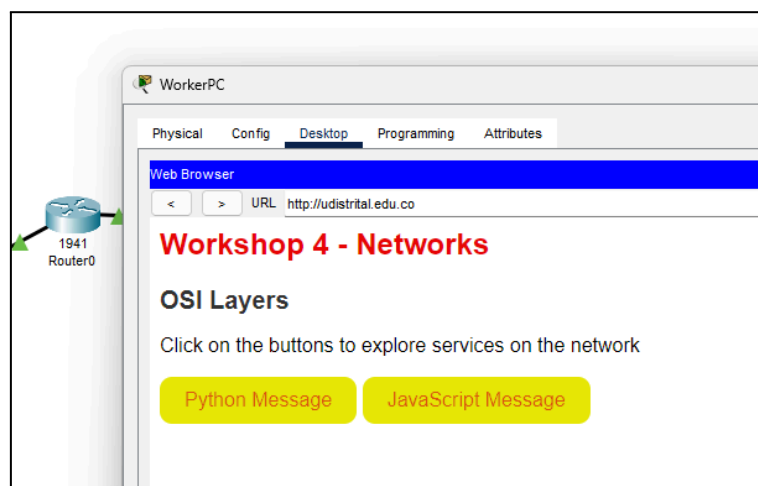


Figure 4 - udistrita.edu.co

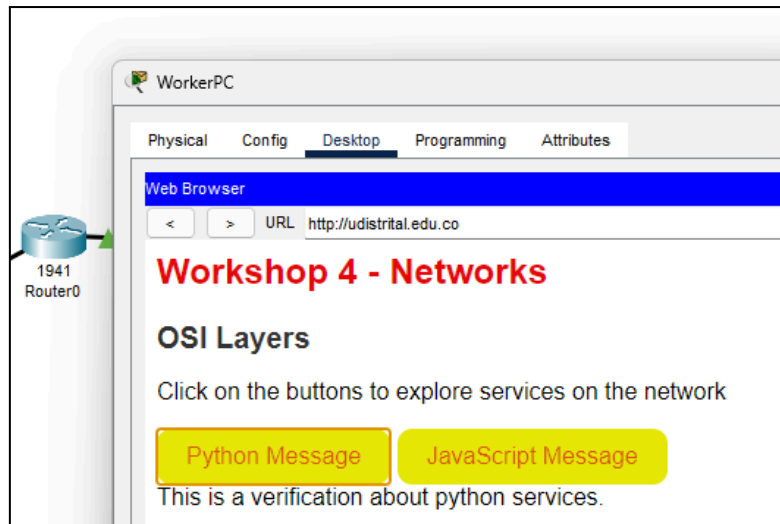


Figure 5 - Python Verification

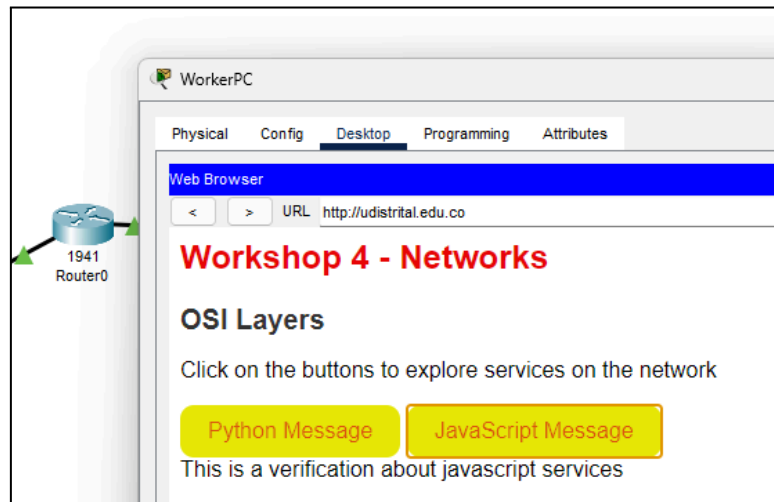


Figure 6 - Javascript Verification

After completing the tests and checking all the configurations in detail, we obtain the final result where the StudentLaptop device can communicate with the server to obtain the page.

This workshop effectively demonstrates the implementation of network services using Python and JavaScript servers in Packet Tracer, showing the interaction between front-end and back-end components through HTTP requests. The hands-on lab reinforces understanding of the OSI model and provides hands-on experience with real-world networking concepts such as IP addressing, routing, and VLAN configuration.